HANDBOOK OF MODULES

Short Description of Mathematics Courses Attended

Submitted in support of the application for admission to

the Master programmes of the Graduate School

“Mathematics as a Key Technology”

RPTU Kaiserslautern-Landau

Programmes Applied For
1. Mathematics for Technology and Data Science *(first preference)*
2. Actuarial and Financial Mathematics
3. Mathematics International

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Date of Completion	16 January 2026
CGPA	3.73 / 4.00 (Total Credits: 134)
Medium of Instruction	English

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1. Introduction and Purpose

This handbook provides a structured description of all mathematics and mathematics-intensive courses that I, Muhammad Anwar, attended and successfully completed during my four-year Bachelor of Science in Software Engineering programme at COMSATS University Islamabad (Sahiwal Campus). The document has been prepared in response to the admission requirement of the Graduate School “Mathematics as a Key Technology” at RPTU Kaiserslautern-Landau, which asks applicants to submit a short description of the mathematics courses they have previously attended, in the form of a handbook of modules.

My undergraduate programme followed the standard Pakistani semester system, with two regular semesters of sixteen weeks each per academic year, supplemented by an optional eight-week summer session. The programme totalled one hundred and thirty-four (134) credit hours and was completed with a cumulative grade point average (CGPA) of **3.73 out of 4.00**. The medium of instruction throughout the programme was English. The descriptions that follow follow the conventions of a European module handbook: each module entry states the course code, title, credit hours, semester of attendance, letter grade achieved, and a short summary of the topics covered, the learning outcomes attained, and the relevance of the module to graduate-level study in mathematics.

The handbook is organised in three parts. Section 2 presents a tabular summary of the six courses carrying an explicit **MTH** prefix, which form the core of my mathematical preparation. Section 3 provides the detailed module descriptions, including two non-credit foundation courses (MTH100 and MTH101) that I completed in addition to the regular degree requirements in order to strengthen my preparation in calculus. Section 4 then describes a set of supporting quantitative and computational courses (Applied Physics, Discrete Structures, Data Structures and Algorithms, Formal Methods, Data Science, Artificial Intelligence, and Computer Vision) whose content is heavily mathematical and which therefore complement the MTH-prefixed modules in building the analytical foundation required for the Master programmes to which I am applying. Section 5 reproduces the official grading scale and the credit-hour definition used by COMSATS University Islamabad.

2. Summary of Mathematics Courses (MTH-prefixed)

The following table lists all six courses with an **MTH** prefix that appear on my official transcript. The two foundation courses, MTH100 (Mathematics I) and MTH101 (Calculus I), were taken on a non-credit (N.C.) basis as supplementary preparation; the remaining four courses carry full credit weight and constitute the formal mathematics component of my undergraduate degree.

Code	Course Title	Cr.	Grade	GP	Semester
MTH100	Mathematics I (Non-Credit Foundation)	N.C.	Pass	–	Preparatory
MTH101	Calculus I (Non-Credit Foundation)	N.C.	Pass	–	Preparatory
MTH104	Calculus and Analytic Geometry	3	A	4.00	Spring 2023
MTH251	Linear Algebra	3	A	4.00	Spring 2023
MTH262	Statistics and Probability Theory	3	A	4.00	Fall 2023

Code	Course Title	Cr.	Grade	GP	Semester
MTH467	Operations Research	3	B+	3.33	Fall 2024
Total	**4 credited MTH courses + 2 non-credit foundations**	**12**	–	**3.78 avg**	**2023–2024**

Grading symbols: A = 4.00 (85–100%), A− = 3.66 (80–84%), B+ = 3.33 (75–79%). Cr. = credit hours; GP = grade points on a 4.00 scale; N.C. = non-credit course taken for supplementary preparation.

3. Detailed Module Descriptions – Mathematics Courses

The following pages contain detailed module-handbook-style descriptions of the six MTH-prefixed courses. Each entry follows the structure recommended for European module handbooks: an identification block (course code, title, credits, grade, semester), a short prose description of the content and learning outcomes, and a bulleted list of the principal topics covered. The two non-credit foundation courses (MTH100, MTH101) are described first, followed by the four credit-bearing courses in chronological order.

MTH100 – Mathematics I (Non-Credit Foundation Course)

Credits: *Non-Credit (N.C.)* | ***Grade:*** *Pass* | ***Grade Points:*** *–* | ***Semester:*** *Preparatory, prior to Spring 2022*

Type: Preparatory mathematics | Format: Lecture + Tutorial | Assessment: Continuous assessment and final examination

Mathematics I was a non-credit preparatory course intended to consolidate the pre-university mathematical background of incoming students before they undertook the formal calculus and linear algebra sequence. The course rehearsed the fundamental operations of algebra, the properties of elementary functions, and the basic vocabulary of trigonometry and coordinate geometry, providing a self-contained bridge between school mathematics and the more rigorous treatment expected at university level. I completed the course successfully on a non-credit basis in order to ensure that my foundation was strong enough to tackle the subsequent credit-bearing calculus sequence with confidence.

Key topics covered:

- Review of algebraic manipulation, equations, and inequalities
- Elementary functions: polynomial, rational, exponential, logarithmic, trigonometric
- Coordinate geometry in two dimensions; lines, circles, conic sections
- Sequences, series, and the binomial theorem
- Introduction to limits and the intuitive notion of continuity

MTH101 – Calculus I (Non-Credit Foundation Course)

Credits: *Non-Credit (N.C.)* | ***Grade:*** *Pass* | ***Grade Points:*** *–* | ***Semester:*** *Preparatory, prior to Spring 2022*

Type: Preparatory calculus | Format: Lecture + Tutorial | Assessment: Continuous assessment and final examination

Calculus I was the second non-credit preparatory course, devoted to a first encounter with single-variable differential and integral calculus. The course introduced the formal ε-δ definition of the limit, the derivative as a limit of difference quotients, and the definite integral as the limit of Riemann sums, together with the fundamental theorem of calculus that links differentiation and integration. Although the course carried no credit weight, it played a decisive role in building the analytical intuition that allowed me to achieve an A grade in the subsequent credit-bearing calculus sequence.

Key topics covered:

- Limits and continuity of functions of one variable
- Differentiation rules; derivatives of elementary functions
- Applications of the derivative: optima, related rates, curve sketching
- The definite integral and the fundamental theorem of calculus
- Techniques of integration and applications to area and volume

MTH104 – Calculus and Analytic Geometry

Credits: *3* | ***Grade:*** *A* | ***Grade Points:*** *4.00* | ***Semester:*** *Spring 2023 (Semester 2)*

Type: Core mathematics | Format: 3 credit hours (3 h lecture / week) | Assessment: Mid-term (50%) + Final examination (50%)

This was the first credit-bearing calculus course of my programme and covered single-variable and multivariable calculus together with the elements of analytic geometry. The course developed the differential and integral calculus of functions of one and several variables, the geometry of curves and surfaces in two and three dimensions, and the basic theory of infinite series. The grade achieved was A, corresponding to a percentage of 85% or above and to a grade point of 4.00. The conceptual maturity acquired in this course underpinned my later work in optimisation, probability, and machine learning.

Key topics covered:

- Functions of several variables; partial derivatives; the chain rule
- Directional derivatives, gradients, and tangent planes
- Multiple integrals (double and triple); change of variables
- Vector calculus: line and surface integrals; the theorems of Green, Gauss, and Stokes
- Infinite sequences and series; power series; Taylor and Maclaurin expansions
- Conic sections and the geometry of quadric surfaces

MTH251 – Linear Algebra

Credits: *3* | ***Grade:*** *A* | ***Grade Points:*** *4.00* | ***Semester:*** *Spring 2023 (Semester 2)*

Type: Core mathematics | Format: 3 credit hours (3 h lecture / week) | Assessment: Mid-term (50%) + Final examination (50%)

Linear Algebra introduced the algebraic structures that pervade modern applied mathematics, statistics, and computational finance. The course covered linear systems, matrices, determinants, vector spaces, linear transformations, eigenvalues and eigenvectors, and inner product spaces, with regular reference to applications in computer graphics, network analysis, and statistical modelling. The course was completed with the grade A (grade point 4.00). The matrix-algebraic way of thinking acquired here is, in my view, the single most important prerequisite for the Actuarial and Financial Mathematics and Mathematics for Technology and Data Science programmes at RPTU.

Key topics covered:

- Linear systems and Gaussian elimination; matrix algebra and inverses
- Determinants and their properties; Cramer’s rule
- Vector spaces and subspaces; linear independence; basis and dimension
- Linear transformations; kernel and image; the rank–nullity theorem
- Eigenvalues, eigenvectors, and diagonalisation; symmetric and orthogonal matrices
- Inner product spaces; Gram–Schmidt orthogonalisation; least-squares approximation

MTH262 – Statistics and Probability Theory

Credits: *3* | ***Grade:*** *A* | ***Grade Points:*** *4.00* | ***Semester:*** *Fall 2023 (Semester 3)*

Type: Core mathematics | Format: 3 credit hours (3 h lecture / week) | Assessment: Mid-term (50%) + Final examination (50%)

This course provided a rigorous introduction to probability theory and mathematical statistics, building on the calculus and linear algebra of the previous year. The probability component covered the axiomatic foundations, discrete and continuous random variables, joint and conditional distributions, and the law of large numbers and the central limit theorem. The statistics component treated point and interval estimation, hypothesis testing, and an introduction to linear regression. The course was completed with the grade A (grade point 4.00). This module is the natural prerequisite for the stochastic processes, risk theory, and financial mathematics courses that form the core of the Actuarial and Financial Mathematics programme at RPTU.

Key topics covered:

- Axioms of probability; conditional probability and Bayes' theorem
- Discrete and continuous random variables; standard distributions
- Joint distributions; covariance, correlation, and independence
- Law of large numbers and the central limit theorem
- Point and interval estimation; properties of estimators
- Hypothesis testing; type I and type II errors; p-values
- Simple and multiple linear regression; analysis of residuals

MTH467 – Operations Research

Credits: *3* | ***Grade:*** *B+* | ***Grade Points:*** *3.33* | ***Semester:*** *Fall 2024 (Semester 5)*

Type: Applied mathematics | Format: 3 credit hours (3 h lecture / week) | Assessment: Mid-term (50%) + Final examination (50%)

Operations Research introduced the principal mathematical models and methods of constrained optimisation, with explicit reference to decision-making in industry, finance, and the public sector. The course covered linear programming and the simplex method, duality and sensitivity analysis, the transportation and assignment problems, integer programming, network optimisation, and an introduction to dynamic programming and queuing models. The course was completed with the grade B+ (grade point 3.33). This module is particularly relevant to the Mathematics for Technology and Data Science programme, in which optimisation plays a central role, and to the Actuarial and Financial Mathematics programme, where optimisation underpins portfolio selection and risk-budget allocation.

Key topics covered:

- Linear programming: formulation, graphical solution, the simplex method
- Duality theory and economic interpretation of dual variables
- Sensitivity and post-optimality analysis
- Transportation, transhipment, and assignment problems
- Network models: shortest path, maximum flow, minimum spanning tree
- Integer and mixed-integer programming; branch-and-bound
- Dynamic programming and Markov decision processes; introduction to queuing theory

4. Supporting Quantitative and Computational Courses

In addition to the six MTH-prefixed courses described above, my undergraduate programme contained a number of courses in physics, theoretical computer science, and data-driven computing whose mathematical content is substantial. These courses developed my fluency with mathematical reasoning in settings outside a pure mathematics classroom and are therefore directly relevant to graduate study in applied mathematics. The descriptions below focus specifically on the mathematical content of each course.

PHY121 – Applied Physics for Engineers

Credits: *4 cr.* | ***Grade:*** *A* | ***Semester:*** *Spring 2022*

Covered mechanics, oscillations and waves, electromagnetism, and an introduction to quantum phenomena, with a strong emphasis on the differential and integral calculus used to formulate physical laws. The course reinforced my fluency with vector calculus, ordinary differential equations, and the modelling of physical systems.

CSC102 – Discrete Structures

Credits: *3 cr.* | ***Grade:*** *A* | ***Semester:*** *Fall 2022*

An introduction to the discrete mathematics that underpins computer science: propositional and predicate logic, set theory, relations and functions, combinatorics, graph theory, and modular arithmetic. The course developed rigorous proof skills and the algebraic vocabulary that recurs in cryptography, coding theory, and algorithm analysis.

CSC211 – Data Structures and Algorithms

Credits: *4 cr.* | ***Grade:*** *A* | ***Semester:*** *Fall 2023*

Studied the design and analysis of algorithms with explicit reference to asymptotic complexity (Big-O, Big-Θ, Big-Ω notation), recurrence relations, and amortised analysis. Mathematical content includes the master theorem, graph algorithms (DFS, BFS, Dijkstra, Kruskal, Prim), dynamic programming, and greedy algorithms.

CSE356 – Formal Methods

Credits: *3 cr.* | ***Grade:*** *A* | ***Semester:*** *Spring 2024*

An introduction to the use of mathematical logic and discrete mathematics for the specification and verification of software systems. Topics included propositional and first-order logic, model checking, temporal logics (LTL and CTL), and the use of formal specification languages. The course required constructing rigorous mathematical proofs and reasoning about program correctness.

CSC461 – Introduction to Data Science

Credits: *3 cr.* | ***Grade:*** *A* | ***Semester:*** *Spring 2024*

Covered the mathematical and computational foundations of data-driven modelling: matrix decomposition (SVD, PCA), linear and logistic regression from a likelihood perspective, cross-validation, regularisation, and the bias–variance trade-off. The course made extensive use of multivariable calculus, linear algebra, and probability theory.

CSC462 – Artificial Intelligence

Credits: *4 cr.* | ***Grade:*** *A−* | ***Semester:*** *Fall 2024*

A mathematically-oriented treatment of artificial intelligence, including search algorithms, constraint satisfaction, Markov decision processes, Bayesian networks, and an introduction to statistical learning. The course applied dynamic programming, probability theory, and linear algebra to questions of inference and decision-making under uncertainty.

CSC455 – Computer Vision

Credits: *3 cr.* | ***Grade:*** *A−* | ***Semester:*** *Fall 2025*

Covered the mathematical foundations of image processing and computer vision, including linear filtering in the spatial and frequency domains, the Fourier transform, edge detection, camera geometry and projective transformations, and the use of singular value decomposition in structure-from-motion. The course relied heavily on linear algebra and multivariable calculus.

5. Grading Scale, Credit-Hour Definition and Academic Standing

The following information is reproduced from the official transcript issued by COMSATS University Islamabad and explains the grading and credit-hour conventions used throughout this handbook.

5.1 Grading Scale

Each course is assessed on a scale of 100 marks, of which 50% are reserved for continuous assessment (mid-term examinations, quizzes, assignments, and laboratory work) and 50% for the terminal examination. The minimum pass mark in any course is 50%. Marks are converted to letter grades and grade points according to the following scale, which is reproduced verbatim from my transcript.

Percentage	Letter Grade	Grade Points	Percentage	Letter Grade	Grade Points
85 and above	A	4.00	61 – 63	C	2.00
80 – 84	A−	3.66	58 – 60	C−	1.66
75 – 79	B+	3.33	54 – 57	D+	1.30
71 – 74	B	3.00	50 – 53	D	1.00
68 – 70	B−	2.66	Below 50	F	0.00
64 – 67	C+	2.33			

5.2 Credit-Hour Definition

Following the standard convention of the Higher Education Commission of Pakistan, one credit hour corresponds to one hour of lecture per week over a sixteen-week semester, or equivalently to three hours of laboratory work per week over the same period. A typical three-credit course therefore represents approximately forty-eight contact hours of instruction in a regular semester, in addition to independent study. The COMSATS semester is sixteen weeks long, with an optional eight-week summer session that follows the same weekly load.

5.3 Academic Standing

Over the course of my four-year Bachelor of Science in Software Engineering programme I registered for a total of **134 credit hours** and passed all of them, accumulating a total of **499.67 credit points** and achieving a **cumulative grade point average (CGPA) of 3.73 out of 4.00**. Within the mathematics component specifically, I obtained the grade A (4.00) in Calculus and Analytic Geometry, Linear Algebra, and Statistics and Probability Theory, and the grade B+ (3.33) in Operations Research, giving an average grade point of approximately 3.83 across the four credit-bearing MTH courses. This record demonstrates a consistent level of academic excellence in mathematics that I intend to build upon at the Graduate School “Mathematics as a Key Technology” of RPTU Kaiserslautern-Landau.

This handbook has been prepared by the applicant on the basis of his official transcript issued by COMSATS University Islamabad, attested by the Higher Education Commission of Pakistan (Regional Centre, Lahore) on 4 May 2026. The original transcript is submitted separately as part of the application package.